Deep Learning Assignment: SMS Spam Detection using Vanilla RNN (SimpleRNN)

Dataset

Use: `spam.csv`
Typical columns:

- `v1` = label (`ham` / `spam`)
- `v2` = message text
  Drop extra empty columns like `Unnamed: 2/3/4` if present.

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1) Learning Outcomes

By the end of this assignment, you should be able to:

- Clean and prepare text data for sequence models (tokenize → sequence → pad)
- Train a **vanilla RNN (SimpleRNN)** model for binary classification
- Evaluate using **precision, recall, F1-score**, and confusion matrix
- Improve performance via tuning and error analysis (without LSTM/GRU)

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2) Problem Statement

Build a model that classifies an SMS as:

- **Spam (1)** or **Ham (0)**

Mandatory constraint

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- Use **vanilla RNN only**: `SimpleRNN` as the main sequence layer.
- Do **not** use: LSTM, GRU, Transformers, attention-based architectures.

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3) Tasks & Deliverables

Task A — Data Cleaning & Exploration (Mandatory)

1. Load the dataset with correct encoding (`latin-1` is commonly needed).
2. Keep only the label and message text columns.
3. Rename columns to `label` and `text`.
4. Encode labels: `ham → 0`, `spam → 1`.
5. Show:
    - dataset shape after cleaning
    - class balance (counts and %)

Deliverable: A short "Data Summary" section with outputs.

Task B — Preprocessing for RNN (Mandatory)

1. Split the dataset into **train/validation/test** (recommended: 80/10/10 with stratification).
2. Tokenize messages at **word-level** (set a vocabulary cap, e.g., 10,000).
3. Convert messages to sequences and **pad** them to a fixed length.
4. Explain your preprocessing decisions:
    - Why this vocabulary size?

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- Why this max sequence length?
- How do you handle unknown words?

Deliverable: “Preprocessing Decisions” section + shapes of padded datasets.

Task C — Build and Train the Vanilla RNN Model (Mandatory)

Build and train a model with:

- **Embedding layer**
- **SimpleRNN layer** (choose units like 32/64/128)
- Dropout (recommended)
- Dense output with sigmoid activation

Training requirements:

- Use `binary_crossentropy`
- Use an optimizer like Adam
- Use **early stopping** (recommended)
- Plot training curves (loss + accuracy)

Deliverable: Model summary + training curves + brief observations.

Task D — Evaluation & Error Analysis (Mandatory)

Evaluate on the **test set** and report:

- Accuracy
- Precision

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- Recall
- F1-score
- Confusion matrix

Error analysis:

1. Show **5 misclassified examples** with:
    - actual label
    - predicted probability
    - predicted label
    - message text (truncate if long)
2. Short reflection (3–5 lines):
    - Which is worse in real life and why?
        - False positive (ham labeled spam)
        - False negative (spam labeled ham)

Deliverable: “Evaluation & Error Analysis” section.

Task E — Improve the Model (Pick Any 2) (Mandatory)

You must run:

- **Baseline model**
- **Improved model** (with 2 enhancements)

Choose any 2 enhancements:

1. Tune max sequence length (e.g., 60 vs 120 vs 200)

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2. Tune RNN units (32 vs 64 vs 128)
3. Adjust dropout / recurrent dropout
4. Add an extra Dense layer (small MLP head)
5. Use class weights to handle imbalance
6. Tune decision threshold (0.5 → other value)

Deliverable: A small comparison table (baseline vs improved) + 2–3 lines explaining what helped.

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4) Submission Format

Submit a single **Jupyter Notebook (.ipynb)** with:

- Markdown explanations between steps
- Output cells showing results
- Plots (training curves)
- Final metrics + confusion matrix
- Misclassification examples
- Baseline vs improved comparison table

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5) Grading Rubric (50 points)

- Data cleaning + exploration: **5**
- Preprocessing correctness + reasoning: **10**
- Vanilla RNN model build + training setup: **10**

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- Evaluation metrics + confusion matrix: **10**
- Error analysis + misclassified examples: **10**
- Improvements (2+) + comparison: **5**

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